

How East-West POWER CORRIDOR Can Extend Time For Solar-PV And Wind Power Generation

Embracing innovative solutions like a dedicated east-west HVDC corridor for combined solar-PV and wind power transmission can pave the way for a more sustainable energy future, blending efficiency with environmental stewardship



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The energy mix of our electric grid is changing very fast. Every year, renewable energy is being added to combat global warming. Solar-PV and wind power are the two major renewable sources. However, both these sources are highly fluctuating in nature. Hence, as their share in the grid increases, the chances of the grid becoming unstable are higher. Therefore, extra measures must be taken to keep the grid stable. Table 1 shows the energy mix.

Hybrid (solar-PV + wind) farms are being established, as solar power generation happens during the day, while wind power is mostly available at night. Such hybrid farms offer a steady output throughout 24 hours. There are plans to establish grid-scale batteries which store energy whenever there is excess generation. The stored energy is used

TABLE 1
INDIA'S INSTALLED POWER GENERATION CAPACITY

Thermal	Nuclear	Renewables			Total	Units
		Hydro	Solar	Wind		
237	7	47	67	42	400	GW
59.25	1.75	11.8	16.8	10.5	100	%

when the demand is more than the generation. Here, we propose a more elaborate scheme, which combines different geographical locations and time zones for establishing solar-PV and wind farms. These farms are interconnected through high voltage DC (HVDC) transmission lines. Such a power corridor offers multiple benefits.

Why East-West

Our country lies between longitudes $68^{\circ} 07' E$ (Gujarat) and $97^{\circ} 25' E$ (Arunachal Pradesh), with an angle difference of $29^{\circ} 18'$. The rotation time of Earth per degree longitude is approximately 4 minutes.

So, rotation time of Earth per degree longitude = $24 \text{ hrs} \times 60 \text{ min} / 360 = 4 \text{ minutes}$.

And the time taken to cover $29^{\circ} 18'$ is given by $(29^{\circ} \times 4) + ((18' / 60) \times 4) = 117.2 \text{ minutes}$.

Thus, the time difference between sunrise in the east to sunrise in the west is almost two hours. This means a solar plant in the east starts generating power two hours before the plant in the west. Similarly, in the evening, the solar power plant continues to generate power for two hours after the plant in the east has stopped its generation. This is shown in Fig. 1. The west-side power genera-

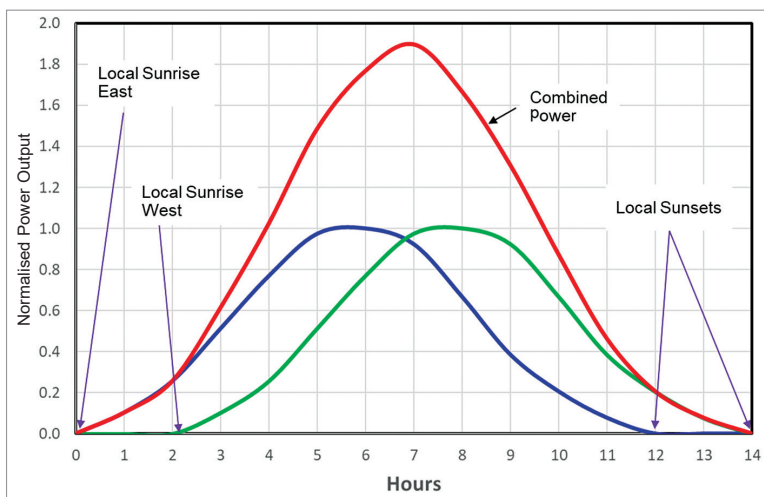


Fig. 1: Plots of solar-PV plant outputs on a sunny day in summer. Blue=Eastern side, Green=Western side, Red=Combined power